

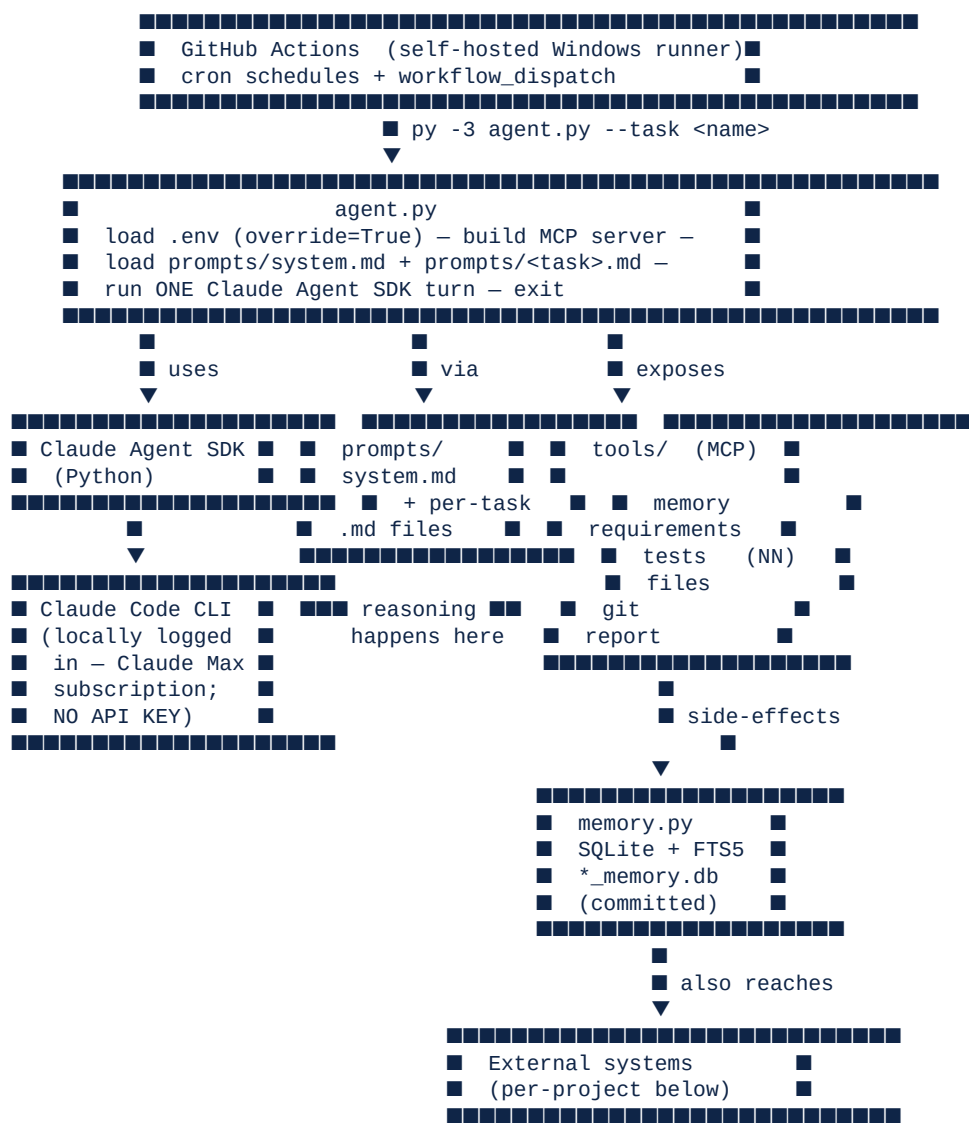
NAPCO Nucleus & Digital Deputy

Architectural Overview

Two Claude-Agent-SDK projects, one architecture. This document captures the layout that NN and DD both follow, then notes where they diverge — by domain, external systems, and tools — so a third agent built to the same pattern can be stood up by swapping only the tool surface and prompts.

1. Shared architecture

Both projects follow the same four-layer shape: a GitHub Actions trigger fires `agent.py`, which loads a prompt, builds an MCP tool server, and runs a single Claude Agent SDK turn. Claude reasons; tools wrap external I/O. Memory is SQLite-with-FTS5, committed to the repo so state travels with `git clone`.



Principles both projects share:

- One `.env` at the project root is the single source of truth for secrets (Digital-Deputy-style).
- One `google-credentials.json` at the project root holds the Google service-account key.

- Claude Code CLI does the LLM work, authenticated via your Claude Max subscription. No `ANTHROPIC_API_KEY` is set anywhere.
- The memory database is committed to the repo, so cloning the project on a new machine instantly recovers prior context.
- Algorithmic work lives in `PROMPTS`, not Python. Tools wrap I/O (call an API, read a file, run a command). Reasoning, classification, summarization — all done by Claude in the prompt.

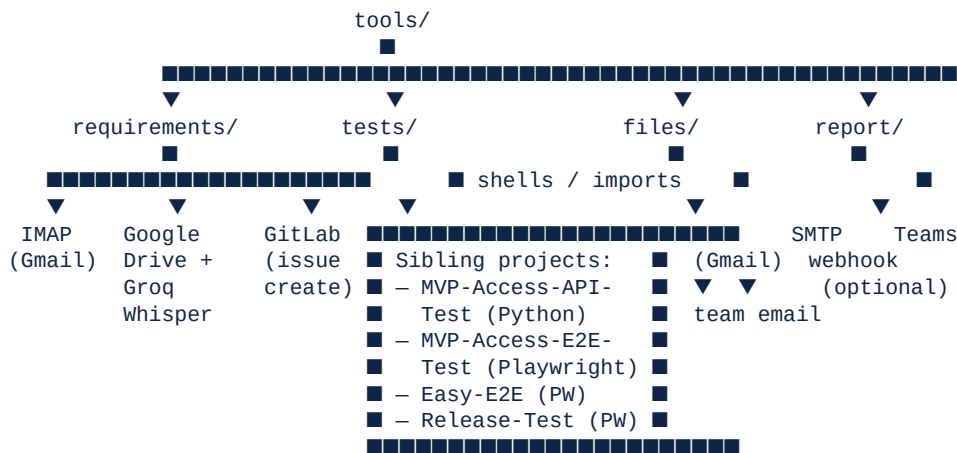
2. Where DD and NN diverge

Same skeleton; different muscles. The table below lists every concern that differs and what each project does instead.

Concern	Digital Deputy (DD)	NAPCO Nucleus (NN)
Domain	Job search + SATUMM business outreach + recruiter tracking	Testing information + project management for MVP Access
Workflows	6: daily-report, find-importers, find-exporters, find-recruiters, job-post, requirements-management	8: daily-report, job-post, requirements-management, api-functional-test, api-integration-test
Prompts	system.md + 5 task prompts	system.md + 6 task prompts
Tools	memory, search, apply, track, guards, report	memory, requirements, tests, files, git, report
Outbound email	SMTP × 2 identities (BUSINESS_ * SATUMM, PERSONAL_ * SATUMM)	SMTP × 1 identity (NAPCO Nucleus <khasan@ael-bd.com>)
Inbound email	IMAP for reply polling	IMAP for requirement intake
Structured data	Google Sheets (sheets.py)	Google Drive folder (recordings + PDFs)
Issue tracker	—	GitLab (gitlab_client.py)
AI services	Claude only (via local CLI)	Claude (CLI) + Groq Whisper (audio→text)
Side-projects	None — DD owns its own work	4 sibling test projects: API-Test (Python import via sys.path), 3× E2E-Test (Python)
Memory DB	deputy_memory.db	nucleus_memory.db
Trigger cadence	Business hours + daily + monthly	Every 2h + daily + weekly

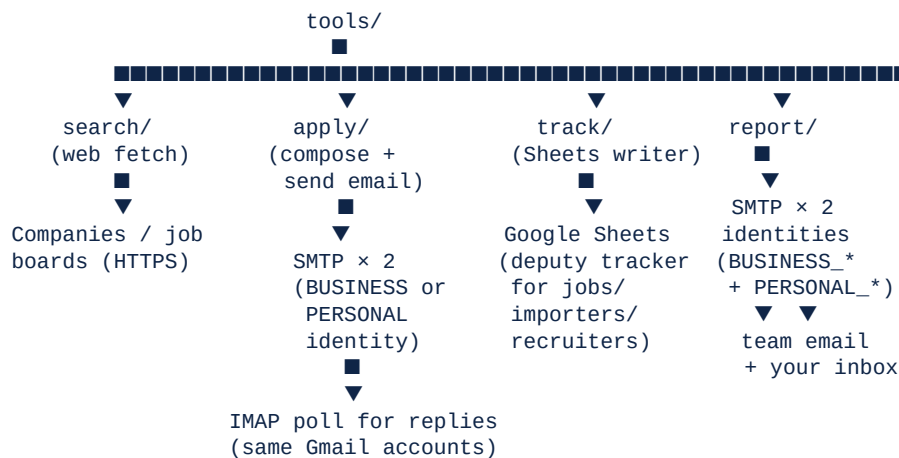
3. NAPCO Nucleus — external dependency map

What NN's six tool submodules actually reach out to. The four sibling test projects are reached two ways: API-Test via Python `sys.path` injection (NN imports its functions and runs them in-process); the three E2E projects via `subprocess.run(["npx", "playwright", "test"])`.



4. Digital Deputy — external dependency map

What DD's tools reach out to. DD's signature is the dual-identity outbound email: the same agent runs as `contact@satumm.com` for trade outreach (`find-importers`, `find-exporters`) and as `titucse@gmail.com` for the personal job search (`find-recruiters`, `job-search`), with separate send-rate caps and identity headers.



5. What this means in practice

Structurally, the two projects are interchangeable. Same auth pattern, same memory pattern, same trigger pattern, same Claude reasoning + Python I/O division. The only project-specific code is the `tools/` submodules and the `prompts/` markdown files. When a third agent comes along, you copy the skeleton and only swap those two folders.

The current refactor (commit 5133ab9) deletes 638 net lines of Python in NN by moving five algorithmic tools (failure RCA classifier, standup formatter, regression diff, test inventory, known-bug list) out of code and into prompts. Claude now reads the same JSON files those tools used to scan and reasons over them directly. Behavior changes by editing markdown, not Python.